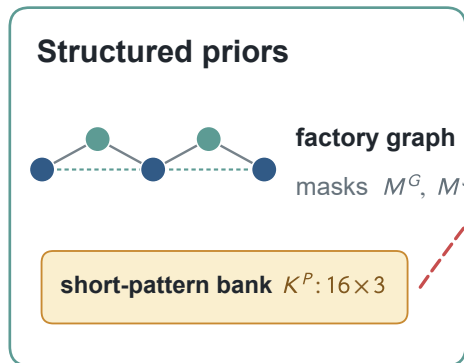
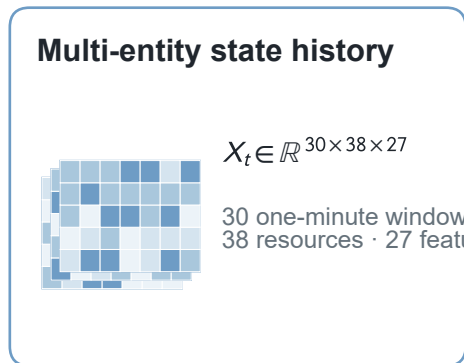
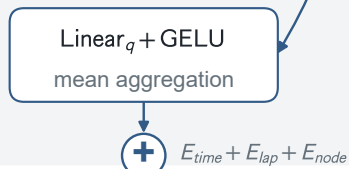
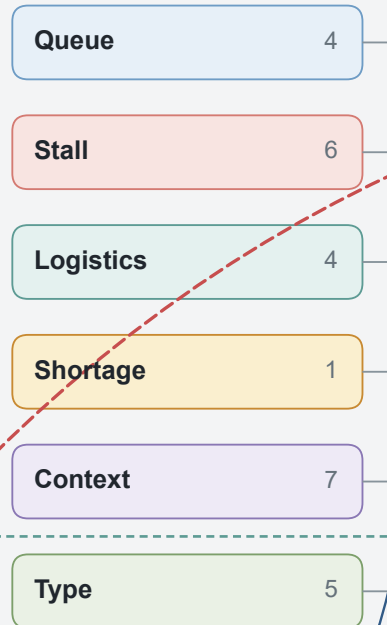


## A Model inputs



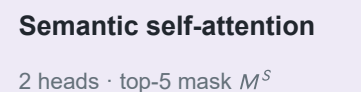
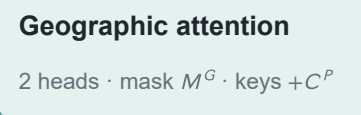
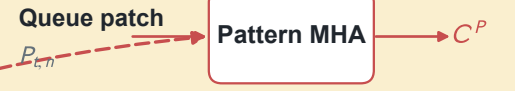
## B Grouped token embedding



**Initial tokens**  $H^{(0)}$

## C Spatiotemporal encoder

pattern keys  $K^P$



Block  $\times 5$  ( $d=80$ )

**Concat + Linear**  
 $[H_T || H_G || H_S]$

**Add & Norm**

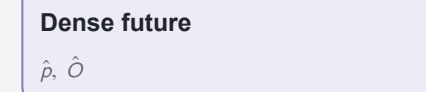
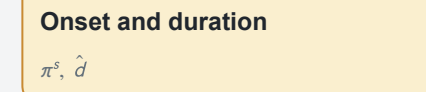
**Feed-forward network**

**Add & Norm**

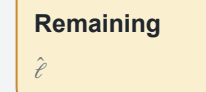
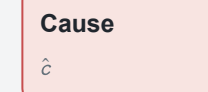
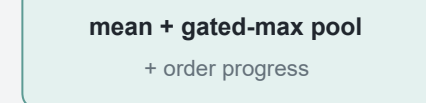
**Encoded states**  $H^{(L)}$

## D Multi-task heads

**Station-level decoding**



**Graph-level decoding**



## E Prediction

**Bottleneck report**

